Concepts of Operating System and Administration

1. Explain Major Functionalities of Operating System.
2. What are different types of CPU Scheduling Algorithms? (FCFS, RR, PRIO, SJF).
3. Explain States of a Process in Operating Systems.
4. Explain Virtual Memory in Operating System. What is Paging/Segmentation?
5. What is difference between process and thread?
6. What is Hyper-V? Explain.
7. What is Active Directory? Explain basic features in Active Directory (AD) of Windows server?
8. What are FSMO roles in Windows Active Directory? How to transfer FSMO roles? What is seize FSMO role option? When to use it?
9. How to take active directory backup? What is authoritative and non-authoritative restore of Active Directory?
10. Explain DNS & DHCP. How to configure in Windows?
11. What is the difference between tree domain vs child domain?
12. What are built in groups - DomainAdmins, Enterprise Admins, Server Operator, Accounts operator, Backup Operator, Domain Users?
13. How to restore a failed Domain controller?
14. Elaborate the concept of Network Policy Server (NPS).
15. What is File Server Resource Manager (FSRM)?
16. What is IIS? Which are the protocols that IIS supports?
17. Explain the concept of Network Load Balancing (NLB).
18. What is PowerShell and what are the features of PowerShell?
19. How to recover forgotten Administrator password in Windows?
20. How to recover forgotten root password in Linux?
21. How to create partition and Filesystem? Write Linux and Windows commands for it.

22. Explain Linux Directory Structure in detail.
23. What is SSH? What is its default port? Why SSH is secure?
24. Explain each field of /etc/passwd. What is /etc/shadow?
25. Why we use LVM in Linux & what are its benefits?
26. Explain Postfix Mail Server. What is default port used for postfix?
27. What Is Dovecot? Can it be used with Postfix?
28. What is mounting in Linux? Write a command to mount FAT formatted pen-drive in /mnt? How can you auto-mount it each time you boot the system?
29. What is hard link and symbolic link? How to create them?
30. What is SELinux?
31. What is NFS? What is root_squash?
32. What is Network Information Service (NIS) in Linux?
33. What is .bashrc and .profile in Linux?
34. Explain Linux architecture. What is name of kernel?
35. What is Kerberos? How does Kerberos work?
36. What is user? What are groups? How to change the logged in user?
37. What is root user and general user? How to create a user? How to add user in group? How to change user's privileges?
38. What is default permission of directory and file? How to change it? What is umask in Linux??

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